

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P2-Model)** **Date: 06-05-2026**

Time: 02.30Pm to 05.30Pm **GTA-30** **Max. Marks: 180**

06-05-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2022-P2)_GTA-30_Syllabus

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2022-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

1. Let $S=\{k\in\mathbb{R}:$ the homogeneous system

$$\left.\begin{aligned} kx+4y+z&=0,\\ 4x+ky+2z&=0, \text{ admits infinitely many solutions}\\ 2x+2y+z&=0\end{aligned}\right\}.$$

Using one or more elements of S (repetition allowed), all possible 2×2 matrices are constructed.

If B denotes such a matrix, then the total number of singular matrices B that can be formed is _____

2. Let $f(x)=ax^2+bx+c$ where a,b,c are integers.

If $\sin\frac{\pi}{7}\sin\frac{3\pi}{7}+\sin\frac{3\pi}{7}\sin\frac{5\pi}{7}+\sin\frac{5\pi}{7}\sin\frac{\pi}{7}=f\left(\cos\frac{\pi}{7}\right)$, then the value of $f(2)$ is

3. The lengths of the major and minor axes of an ellipse are 10 and $2\sqrt{7}$, respectively. A parabola has its vertex at one end of the minor axis of the ellipse and passes through the foci of the ellipse. The parabola and ellipse intersect at two points other than the vertex of the parabola. The distance between these two points is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find n

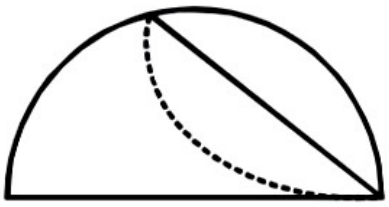
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4. Let $f(x)$ be a polynomial function such that $f(x)>0, \forall\, x \in R$, satisfies the equation $\log_3(f(x)) = \log_3\left(4+\frac{4}{5}+\frac{4}{5^2}+\ldots\infty\right).\log_5\left(1+\frac{f(x)}{f\left(\frac{1}{x}\right)}\right) \forall\, x \in R-\{0\}$, and $f(10)=101$, then the greatest integer value of n for which $\sum_{r=1}^{n} f(r) \le f(10)$, is

5. Let α, β be two distinct non-real roots of the quadratic $ax^2+bx+c=0$ where $a, b, c \in \mathbb{R}$. If $\frac{\alpha^2}{\beta} \in \mathbb{R}$, then the value of $\left|\frac{2\alpha}{\beta}+1\right|^2$ is _____.

6. Three distinct real numbers x_1, x_2, and x_3 in the interval $[0, \pi]$ satisfy the equation $\sec(2x)-\sec x=2$. There are relatively prime positive integers m and n such that $\frac{\pi}{x_1+x_2+x_3}=\frac{m}{n}$. Find $n-m$

7. A semicircular paper is folded along a chord such that the folded circular arc is tangent to the diameter of the semicircle. The radius of the semicircle is 4 units and the point of tangency divides the diameter in the ratio $7:1$. If the length of the crease (the dotted line segment in the figure) is l then determine unit digit of l^2.



8. Let the greatest possible area of the triangle in the complex plane with vertices $2z$, $(1+i)z$, and $(1-i)z$, where z is a complex number satisfying $|4z-2|=1$ is $\frac{a}{b}$(where a, b are coprime), then find $b-a$.

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

9. Let $X=\{1,2,3,4,5,6\}$ and a,c are natural numbers selected from set X with replacement.

Let $f(x)=\begin{cases}x^2+x, & x\le 1\\ x^2-ax, & x>1\end{cases}, g(x)=\begin{cases}x^2-cx, & x\le 2\\ x^2-4x, & x>2\end{cases}$ and $h(x)=f(x).\ g(x), x\in R$.

$N_1=$ Number of function $h(x)$ such that $f(x)$ and $g(x)$ are both discontinuous.

$N_2=$ Number of function $h(x)$ such that $h(x)$ is continuous at $x=1$ and discontinuous at $x=2$ but $f(x)$ and $g(x)$ are both discontinuous.

$N_3=$ Number of function $h(x)$ such that $h(x)$ is discontinuous at $x=1$ and continuous at $x=2$.

Which of the following options is/are true?

A) $N_1=30$ **B)** $N_3=11$ **C)** $N_2=4$ **D)** $N_1+N_2+N_3=45$

10. A tetrahedron ABCD having its faces ABD, ACD, ABC are along the plane $x+y=2, x+z=2, x+y+z=4$ respectively. If $AB=\sqrt{2}, AD=\sqrt{3}, AC=2\sqrt{2}$ where x-coordinate of points B, C, D, are positive then

A) Volume of tetrahedron ABCD is $1/3$

B) Area of plane BCD is $\frac{\sqrt{5}}{2}$

C) Area of plane BCD is $\frac{1}{\sqrt{2}}$

D) Volume of tetrahedron ABCD is $\frac{1}{6}$

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11. Let $f, g:\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ be defined by $f(x) = [2\sin x]\cos\left(x - \frac{1}{2}\right)$ and $g(x) = [5x]\left(x - \frac{1}{5}\right)$, where $[\cdot]$ is the greatest integer function. Which of the following is/are CORRECT?

A) $f + g$ is discontinuous at exactly 14 points in its domain

B) $f + g$ has exactly one point where it is continuous but not differentiable

C) The equation $f(x) = \left(x - \frac{1}{2}\right)^2$ has exactly two real solutions

D) $f'(x) = 0$ has infinitely many solutions

12. The polynomial $(z+i)(z+2i)(z+3i)+10$ has three roots in the complex plane, where $i = \sqrt{-1}$. Which of the following options is/are correct?

A) Area of the triangle formed by these roots is 6

B) Centroid of the triangle formed by the roots of given polynomial has coordinates $(0, -2)$

C) Triangle formed by the roots of given polynomial is equilateral

D) Triangle formed by the roots of given polynomial is right angled triangle.

13. Let A, B and C be $n \times n$ matrices with complex entries satisfying $A^2 = B^2 = C^2$ and $B^3 = ABC + 2I$.

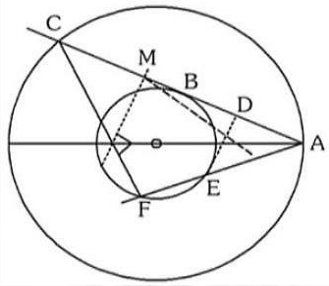
Which of the following options is/are CORRECT.

A) $A^6 = I$ **B)** $C = -A$

C) A is non singular matrix **D)** $B^6 = 6I$

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14. Consider circle $S_1 = x^2 + y^2 - 16 = 0$ and $S_2 = x^2 + y^2 - 25 = 0$. From a point $A(5,0)$ tangent is drawn to $S_1 = 0$ touching it at point B in 1st quadrant and meeting $S_2 = 0$ in point C. Let D be the mid point of AB. A line through point A cuts $S_1 = 0$ in points E and F; such that perpendicular bisectors of DE and CF meets at point M on AC (as shown in figure). Now which of the following options is/are TRUE.



A) $AE \cdot AF = 9$

B) Coordinate of point C is $\left(\frac{7}{5}, \frac{24}{5}\right)$

C) $AD \cdot AC = 18$

D) Coordinate of point $M = \left(\frac{55}{20}, 3\right)$

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SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

15. A set of numbers is called *sum-free* if whenever x and y are(not necessarily distinct) elements of the set, $x + y$ is not an element of the set. For example, $\{1, 4, 6\}$ and the empty set are sum-free, but $\{1, 4, 5\}$ is not. What is the greatest possible number of elements in a sum-free subset of $\{1, 2, 3, ...20\}$?

A) 8 **B)** 9 **C)** 10 **D)** 11

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16. A frog hops along the number line according to the following rules:

- It starts at 0
- If it is at 0, then it moves to 1 with probability $\frac{1}{2}$ and disappears with probability $\frac{1}{2}$
- For $n=1, 2, 3,$ if it is at n, then it moves to $n+1$ with probability $\frac{1}{4}$, to $n-1$ with probability $\frac{1}{4}$, and disappears with probability $\frac{1}{2}$.

What is the probability that the frog reaches 4?

A) $\frac{1}{101}$ **B)** $\frac{1}{99}$ **C)** $\frac{1}{98}$ **D)** $\frac{1}{97}$

17. $A(3,-1,-1)$ and $B(3,1,0)$ are the points in the planes $x+y+z=1$ and $2x-y-z=5$ respectively. C is a variable point lying in both the planes such that the perimeter of ΔABC is minimum. Find the point C.

A) $\left(2,\frac{-1}{\sqrt{2}\left(1+\sqrt{2}\right)},\frac{-1}{\sqrt{2}}\right)$ **B)** $\left(3,\frac{1}{\sqrt{2}\left(1+\sqrt{2}\right)},\frac{1}{\sqrt{2}}\right)$

C) $\left(2,\frac{1}{\sqrt{2}\left(1+\sqrt{2}\right)},\frac{1}{\sqrt{2}}\right)$ **D)** $\left(2,\frac{1}{\sqrt{2}\left(1+\sqrt{2}\right)},\frac{1}{2}\right)$

18. Let f be a function defined on the set of non-negative integers and taking values in the same set. Given that

(a) $x-f(x)=19\left[\frac{x}{19}\right]-90\left[\frac{f(x)}{90}\right]$ for all non-negative integer x; ([.] is G.I.F)

(b) $1900<f(1990)<2000,$

Find the sum of possible values that $f(1990)$ can take

A) 1904 **B)** 3898 **C)** 3984 **D)** 3894

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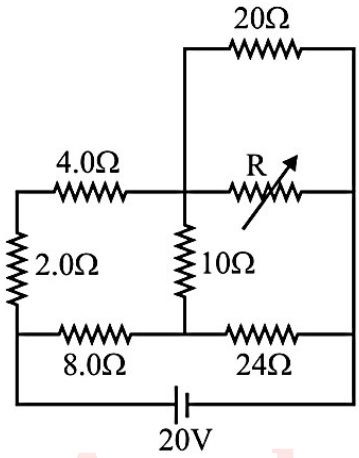
PHYSICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

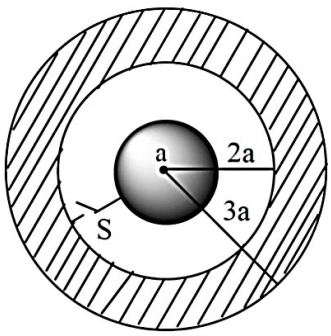
- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

19. For the circuit shown in figure. The value of the variable resistor R (in Ω) in order to minimise the power in the 10Ω resistor is x. Fill the value of $\frac{x}{20}$.

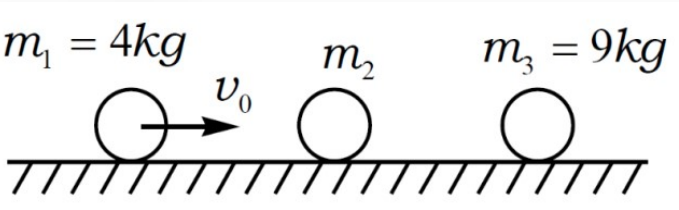


20. A solid conducting sphere of radius 'a' having charge q is surrounded by a concentric conducting spherical shell of inner radius 2a and outer radius 3a as shown in figure. The net charge on spherical shell is $-2q$. If the amount of heat produced when switch is closed is $x\left(\frac{q^2}{96\pi\varepsilon_0 a}\right)$. Fill the value of x.

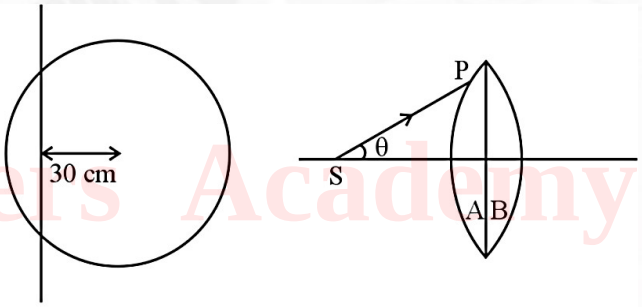


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21. Three balls are kept on a straight line on a smooth surface. The first ball is projected towards middle ball as shown. It collides with middle ball of mass m_2 which then collides with m_3. We want m_3 to move with maximum possible velocity after one collision. What should be mass m_2 (in kg)? All collisions are head on and elastic.



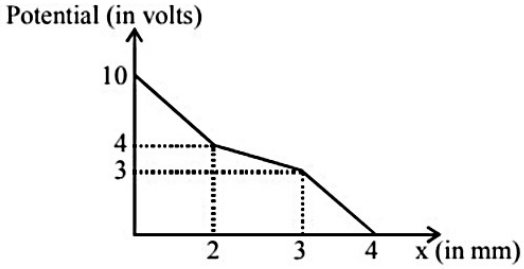
22. There are two spheres of transparent materials. Sphere A has refractive index of $\mu = \frac{8}{5}$ and refractive index of sphere B is not known to us (consider it as μ_2). Radii of each sphere is 50 cm. Both the spheres are cut so that the section of cut is 30 cm from center. Later the smaller parts struck together as shown.

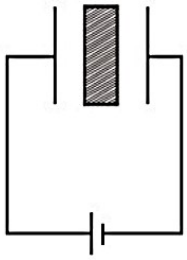


A narrow laser beam from point S (source) is incident on the curved surface of the part of sphere A at point P. The length of beam from S to P is 50 cm. Also the angle made by beam with the principal axis is $\theta = 37^\circ \left(\sin\theta = \frac{3}{5} \right)$. The beam finally emerges from the curved surface of the part of sphere B. The beam follows a total deviation of 53°. If the image of first refraction is formed at a distance D cm from the surface of A, find the value of $\frac{1}{D}$.

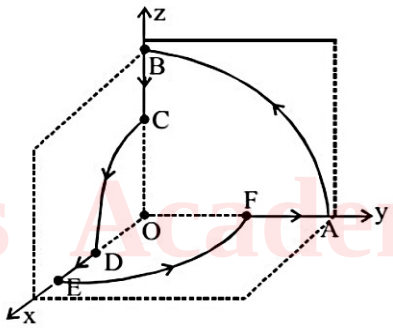
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23. A capacitor filled partially with dielectric material of dielectric constant 'k'. Its electric potential v/s position graph is as shown. Distance between the two plates is 4 mm. Find the dielectric constant of medium.

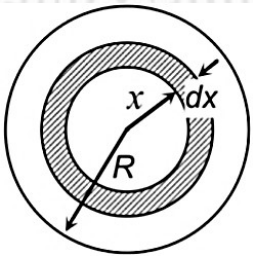




24. In the rigid wire frame ABCDEF, shown in figure, a current $I_0 = 4A$ flows through the wire from F to A and radii of curves AB, CD and EF are a, b and c respectively with centres at O. If this loop is immersed in a uniform magnetic field of magnitude 1.3 T and directed along the line joining (2, 3, 4) to (-7, 2, 0) and allowed to rotate about the x-axis only, then find the value of torque τ (N-m) exerted on the loop by the axis.
(Given: $a = 3m, b = 1m, c = 2m$).

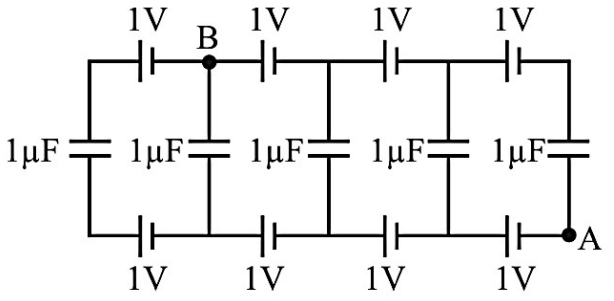


25. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in $N-m$) acting on the disc after 15 sec. (Take $4kQ = 5$ S.I. unit and $R = 1m$).



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26. The potential difference $V_B - V_A$ (in V) for the circuit shown in the figure is



SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

27. The air column in a pipe closed at one end is made to vibrate in its third overtone by a tuning fork of frequency 660 Hz. The velocity of sound in air is 330 m/s. If P_0 represents the mean pressure and ΔP_0 the maximum amplitude of pressure variation in the pipe. Neglect the end correction. Then choose the correct option(s).



A) The length of the pipe is $\ell = \frac{7}{8}$ m

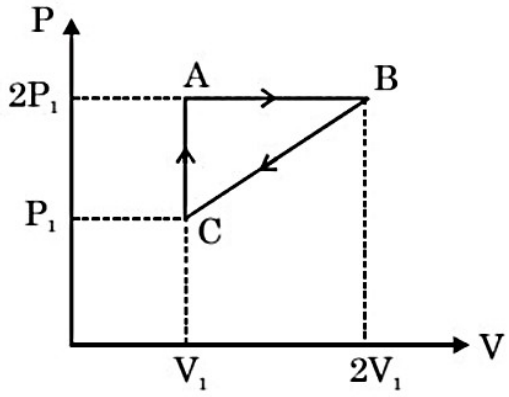
B) The maximum pressure at $x = \frac{\ell}{3}$ will be $\left(P_0 + \frac{\Delta P_0}{2}\right)$

C) The maximum pressure at $x = \frac{\ell}{3}$ will be $\left(P_0 + \frac{\sqrt{3}\Delta P_0}{2}\right)$

D) The minimum pressure at $x = \frac{\ell}{2}$ will be $\left(P_0 - \frac{\Delta P_0}{\sqrt{2}}\right)$

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28. Consider the PV cycle shown. If the heat given to one mole of the monatomic gas in the step AB is Q, then choose correct option(s)



A) Work done in the process BC on the gas is $3Q/10$

B) The heat released in the process BC is $2Q/5$

C) The efficiency of the cycle is 10%

D) The heat given to the gas during CA is $3Q/10$

29. An air filled capacitor has a separation of 0.885 ± 0.009 mm. The area of the plates is $0.040 \pm 0.001\ m^2$. It is connected to a voltage source of voltage 10.0 ± 0.1V. Choose the correct statement(s)

A) The charge has a percentage error of approximately 4.5%.

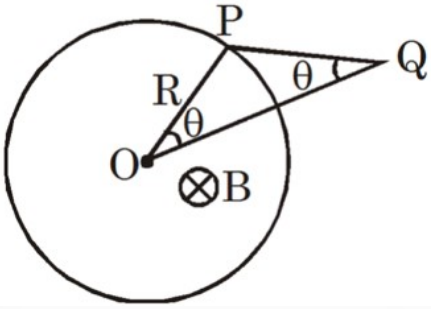
B) The energy stored has a percentage error of approximately 5.5%

C) The force between the plates has a percentage error of approximately 5.5%.

D) If we insert a dielectric of dielectric constant $k = 4$ which fills the region between them completely, the random error in energy increases.

30. In a circular region centered at O and having radius R, magnetic field increases at a constant rate a_0 Tesla/second. A wire frame OPQ having resistance per unit length $\lambda\Omega/m$ is placed as shown in the figure. Choose the CORRECT option(s).

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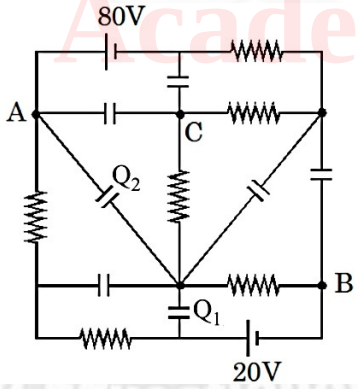
A) $V_P = V_0$ for any θ as there is no induced emf across OP.

B) $V_O - V_Q = \frac{a_0 \pi R^2}{8\left(\sqrt{2}+1\right)}$ for $\theta = 45^\circ$

C) $V_O - V_Q = \frac{a_0 \pi R^2}{2\left(\sqrt{2}-1\right)}$ for $\theta = 45^\circ$

D) $V_P - V_Q = \frac{a_0 \pi R^2}{8\sqrt{2}}$ for $\theta = 45^\circ$

31. For the given circuit, choose the CORRECT statements: (given all capacitors have capacitance $= 1$ F & all resistors have resistance $= 10\Omega$)



A) Potential difference between point A & B $(V_A - V_B)$ is 40 V.

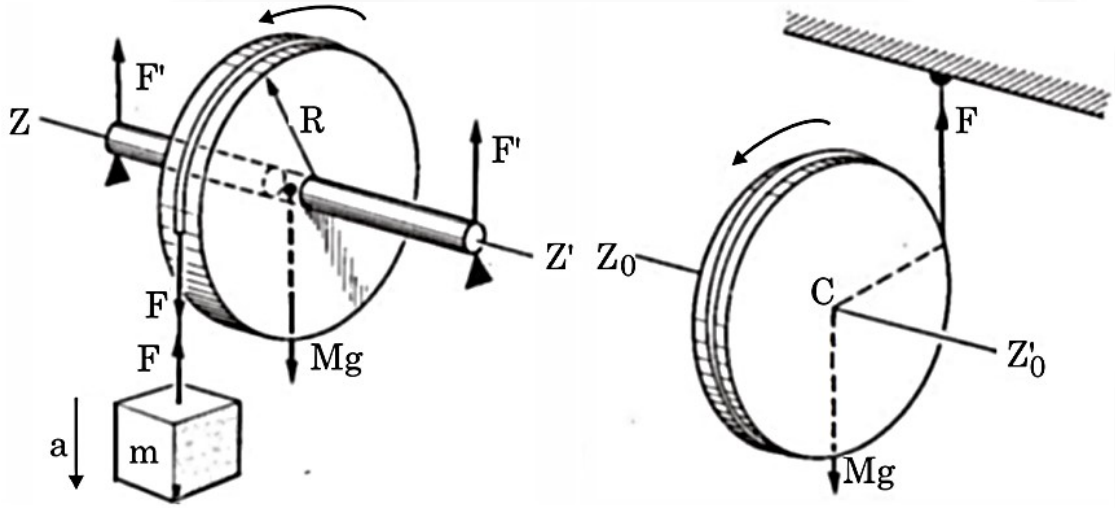
B) Potential difference between point A & B $(V_A - V_B)$ is 50 V.

C) Magnitude of charge Q_1 is 30 C.

D) Magnitude of charge Q_2 is 50 C.

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32. Figure 1 shows the block of mass m connected to a disc of mass M as shown. The disc can rotate freely about its massless axle and the string does not slip on the disc. Figure 2 shows the same disc hanging from an inextensible thread. The string does not slip on the disc. The systems are released from rest simultaneously. The tension in both the strings is F as shown.



A) $m = M$

B) In time *t*, kinetic energy of disc in both the cases is the same

C) In time *t*, the rotational kinetic energy of both the discs is same

D) The angular acceleration of disc is same in both the cases.

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. One half of a rod of length $\ell = 100\,\text{cm}$ has density $\rho_1 = 0.50\ \text{g/cm}^3$ and the other half has density $\rho_2 = 2.50\ \text{g/cm}^3$. The rod is placed horizontally on the horizontal bottom of a large tub and then water is gradually poured in the tub. Find the depth of water in the tub, when the rod makes an angle $\theta = 53^\circ$ with the horizontal. The density of water is $\rho = 1\ \text{g/cm}^3$.

A) 20 cm **B)** 40 cm **C)** 60 cm **D)** 80 cm

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34. In figure 1, we see a ball C colliding against a ball B which is connected to a ball A by a loose inextensible string. There is no friction anywhere. In figure 2, the ball C collides against ball B which is connected to ball A by a taut inextensible string. If the collision is elastic in each of the cases,

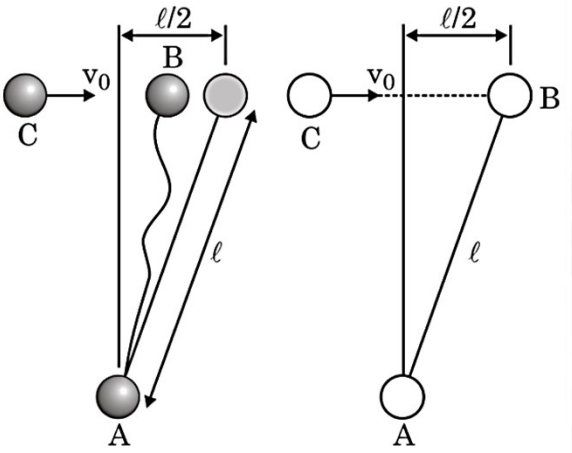


Fig.(1) Fig.(2)

A) The mechanical energy is conserved during and after the collision in both the cases.
B) The mechanical energy is conserved during and after the collision in case1but not in 2.
C)The mechanical energy is conserved during and after the collision in case2 but not in 1.
D) The mechanical energy is not conserved during and after the collision in both the cases.

35. The pitch of screw gauge is 1 mm and its circular scale is divided into 100 divisions. When nothing is put between the studs the circular scale lies between 2nd and 3rd division of main scale, but when circular scale is rotated by 270° the 3rd division of main scale is seen and the 3rd division of main scale coincide with the 12th division of circular scale. When a glass plate is placed between the studs, the circular scale lies between 28th to 29th division of main scale and circular scale reads 20 divisions. Find the thickness of the glass plate in millimeter. (Assume value of one main scale division is same as pitch)

A) 25.83 mm **B)** 26.83 mm **C)** 25.73 mm **D)** 25.80 mm

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36. Abhinav wants to float a ball (perfectly reflecting) in the air. He turned on a monochromatic light source at the bottom, perpendicular to the ground, pointing upwards with intensity P. The ball is dropped from a height H, directly into the light beam. The ball has a radius R and a mass m. Unfortunately, Abhinav did not compute the needed intensity correctly, and the light beam cannot keep it floating in the air. At what speed does the ball hit the ground? (Symbols have their usual meaning.)

A) $\sqrt{H\left(g-\frac{\pi R^2 P}{mc}\right)}$

B) $\sqrt{2H\left(g-\frac{\pi R^2 P}{2mc}\right)}$

C) $2\sqrt{H\left(g-\frac{\pi R^2 P}{mc}\right)}$

D) $\sqrt{2H\left(g-\frac{\pi R^2 P}{mc}\right)}$

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CHEMISTRY **Max Marks: 60**

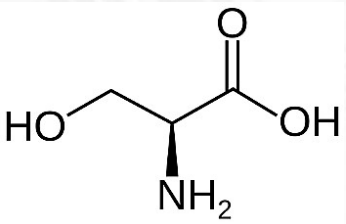
SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

37. If 1.8 g CH_4 contains 0.8 mole neutrons, the number of protons present in 2.0 g CH_2D_2 (made by substituting two hydrogens with deuteriums in the above methane molecule) are _____ moles.

38. If the one letter code of the α – amino acid shown below is followed by each lowercase letter of the English alphabet (a to z) to form two-letter combinations, how many of these 26 combinations would correspond to symbols of elements in the modern periodic table.



39. One mole of acetone reacts with excess thionyl chloride and the only gaseous product evolved is collected in flask–1. One mole of aluminium carbide reacts with excess water and the only gaseous product evolved is collected in flask–2. The two flasks have equal volume and are maintained at the same temperature. If pinholes of areas 2A and A are made in flask–1 and flask–2 respectively, the ratio of initial rates of effusion (r_2/r_1) is_____

40. 1 mole of a vicinal diol on Malaprade oxidation using H_5IO_6 furnishes 1 mole each of the simplest aldehyde and the simplest ketone. Report the sum of all locants in the IUPAC name of the vicinal diol.

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41. A fixed volume of 0.1 M benzoic acid (pKa = 4.199) solution is added to 0.2 M sodium benzoate solution to make a total 300 ml of an acidic buffer. If pH of this buffer solution is 4.5, in what volume ratio of salt to acid were the solutions mixed.

42. *Butanal* can be converted to *butanone* in minimum 4 sequential steps as shown below

$$\text{Butanal} \xrightarrow[ii)\ H_2O]{i)\ NaBH_4} \xrightarrow[\Delta]{c.H_2SO_4} \xrightarrow{H_3O^+(\text{dilute})} \xrightarrow{P.C.C} \text{Butanone}$$

What is the minimum number of steps in which the above conversion can be reversed (i.e. butanone can be converted to butanal) if the following are prohibited.

i) use of carbon chain extending reagents like Grignard reagent that add carbon atoms from outside

ii) use of any dehydrating agent

Note:

1. A single step is defined as one synthetic operation where the substrate is taken, treated with one set of reaction conditions to obtain and isolate the next product in the sequence.

2. The reactions that inherently need sequential additions are treated as a single step only. For example; $NaBH_4$ followed by H_2O (shown above), O_3 followed by Zn/H_2O during reductive ozonolysis of alkenes etc.

3. In case multiple products are possible in a particular step, consider the structure of the major product only.

43. How many of the following 9 statements about nitric acid (HNO_3) is/are correct

1. Pure nitric acid is a colourless liquid but on exposure to light it turns slight brown

2. It can be prepared in the laboratory by distilling a mixture of $NaNO_3$ and concentrated H_2SO_4 at elevated temperature in a glass retort.

3. The boiling point of pure HNO_3 is lesser than the boiling point of pure H_2O at 1 atm. atmospheric pressure.

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4. Lab grade nitric acid contains nearly 68% HNO_3 and 32% H_2O by mass and shows negative deviation from Raoult's law.

5. The boiling point of lab grade nitric acid is greater than the boiling point of pure H_2O at 1 atm. atmospheric pressure.

6. The pKa value of HNO_3 at $25^\circ C$ is below zero.

7. It is used for *picking* of stainless steel.

8. It destroys optical activity of D-Erythrose

9. It is called *aqua fortis* (strong water) due to its ability to dissolve many metals including the less reactive ones like copper and silver.

44. When the monomer of addition polymer - *Teflon* is fluorinated in dark, it forms a stable and saturated compound $-\boldsymbol{Y}$ which when mixed with XeF_2 under plasma conditions converts to an unstable waxy white solid $-\boldsymbol{Z}$ along with the formation of a gaseous by-product. C_2H_6 reacts with excess of this gaseous by-product in u.v. light to form $\boldsymbol{Y}$ via a similar reaction mechanism which operated in the formation of $\boldsymbol{Z}$ under plasma conditions.

If

$a =$ maximum number of colinear atoms in $\boldsymbol{Z}$

$b =$ maximum number of coplanar atoms in $\boldsymbol{Y}$

$c =$ total number of lone pair of electrons in $\boldsymbol{Z}$

$d =$ maximum number of coplanar atoms in the monomer of *Teflon*.

Report the value of $\sqrt{(b+c)-(a+d)}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

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45. The mechanism of S_{N_1} reaction is given as

$$R{-}X \longrightarrow \underset{\text{ion}-\text{pair}}{R^{\oplus}\ X^{\ominus}} \longrightarrow \underset{\text{Solvent separated ion-pair}}{R^{\oplus} \parallel X^{\ominus}} \xrightarrow{Y^{\ominus}} R{-}Y + X^{\ominus}$$

A student writes the following conclusions based on the above mechanism. Identify the one(s) which is/are correct

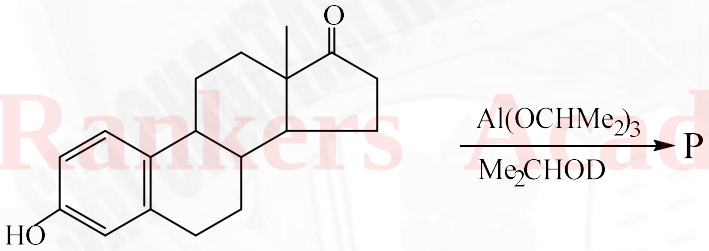
A) $R^{\oplus}$ would be easily formed if substituents are alkyl groups instead of hydrogen atoms.

B) high dielectric constant of solvent enables more efficient separation of ions.

C) diethyl ether is a preferred solvent over isomeric 1-butanol in this reaction.

D) R-Y formed is always a perfect racemic mixture.

46.



Identify the correct statement(s)

A) the structure of P contains 5 chiral carbons

B) P when treated with a big chunk of Na metal evolves D_2 gas

C) P when treated with few drops of CH_3MgBr in ether evolves CH_4 gas

D) P is practically insoluble in water

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47. In the following chart showing atomic and physical properties of the alkali metals, some fields related to sodium and potassium have been concealed by alphabets A to H.

Property	Lithium Li	Sodium Na	Potassium K	Rubidium Rb	Caesium Cs
Atomic number	3	11	19	37	55
Atomic mass (g mol^{-1})	6.94	22.99	39.10	85.47	132.91
Electronic configuration	[He] $2s^1$	[Ne] $3s^1$	[Ar] $4s^1$	[Kr] $5s^1$	[Xe] $6s^1$
Ionization enthalpy / kJ mol^{-1}	520	A	B	403	376
Hydration enthalpy/kJ mol^{-1}	–506	C	D	–310	–276
Metallic radius / pm	152	186	227	248	265
Ionic radius M^+ / pm	76	102	138	152	167
m.p. / K	454	E	F	312	302
b.p / K	1615	1156	1032	961	944
Density / g cm^{-3}	0.53	G	H	1.53	1.90
Standard potentials $E^{\ominus}$/ V for (M^+ / M)	–3.04	–2.714	–2.925	–2.930	–2.927

Which of the following is true (magnitude wise)

A) $A > B$ **B)** $C > D$ **C)** $E > F$ **D)** $G > H$

48. Consider the following 2 complexes X & Y

X : $[Co(NH_3)_3(NO_2)_3]$

Y : $[Co(NH_3)_6][(Co(NO_2)_6]$

Identify the correct statement(s):

A) X can exist as 2 geometrical isomers, both of which are optically inactive

B) If the neutral ligands in X are replaced by 'dien' ligand, the fac - isomer will contain a POS passing through secondary nitrogen atom

C) If all neutral ligands in Y are exchanged with en ligands and half the anionic ligands in Y are exchanged with their linkage counterpart, a total of 4 different stereoisomers can be made

D) X and Y are polymerisation isomers

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49. Which of the following is a preferred quality of compound used as salt bridge:

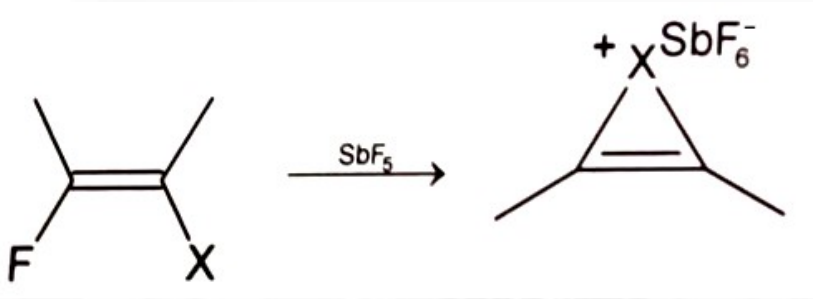
A) it should contain only ionic bonds

B) it should offer robust and near equal cation & anion mobility in aqueous solution

C) it should not react with (precipitate) the electrolytes present in the connected half cells

D) it should be soluble in the ionic agar-agar gel matrix

50.



X = halogen atom other than fluorine

Which of the following factors explain the above reaction :

A) F^- is the best leaving group among halide ions

B) SbF_5 is a voracious Lewis acid with extreme affinity for fluoride ions.

C) the *cyclohalonium* cation formed is aromatic

D) C – F bond is weaker than C – X bond

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

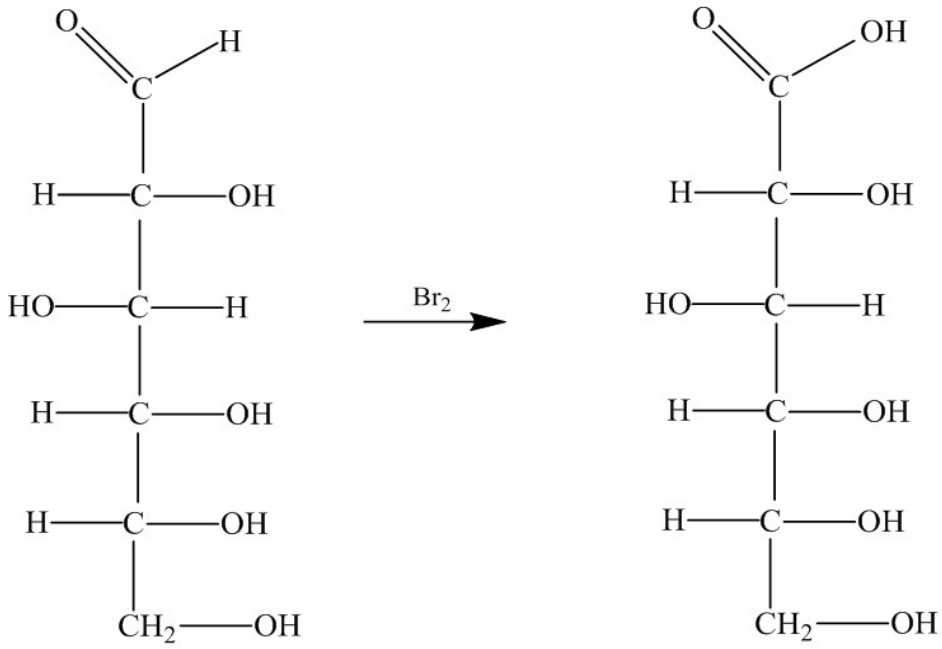
This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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51.



When 1 mole glucose reacts with 3 moles Br_2, it converts to gluconic acid as shown above. In this reaction Br_2 appears in the product as:

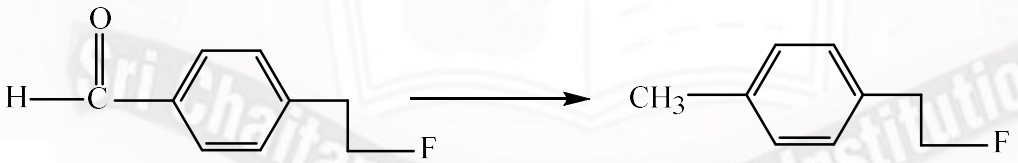
A) 2 moles BrO^- and 2 moles unreacted Br_2

B) 4 moles Br^- and 1 mole unreacted Br_2

C) 6 moles Br^-

D) 2 moles Br_3^-

52. Consider the following conversion

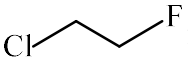


Identify the correct statement

A) it can be achieved by Clemmensen reduction

B) it can be achieved by Wolff Kishner reduction

C) it can be achieved by Red P/HI reduction

D) the reactant molecule can be made by treating benzaldehyde with 

in the presence of anhydrous BF_3

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53. When Dolomite mineral is dissolved in mineral acid, the following reaction takes place

$$\underset{\text{(Dolomite)}}{CaCO_3 \cdot MgCO_3} + 4HCl \longrightarrow CaCl_2 + MgCl_2 + 2H_2O + 2CO_2$$

Identify the correct statement if the atomic weight of each element in Dolomite is double of its atomic number

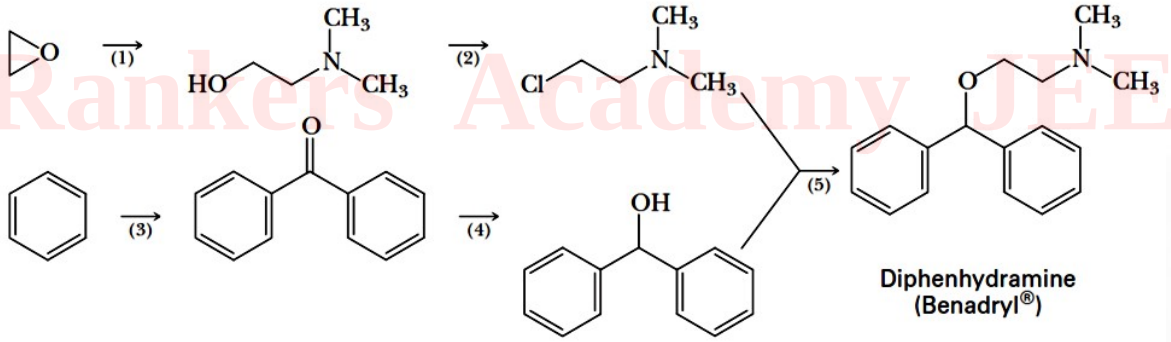
A) EW (equivalent weight) of dolomite is 92.

B) the VD (vapour density) of the only gas evolved in the above reaction is 22 at all temperature and pressure that permit ideal gas behaviour

C) a white precipitate is formed as product.

D) passing the gas evolved in the above reaction through lime water regenerates dolomite mineral.

54. The following flowchart shows the synthesis of diphenylhydramine - a compound whose hydrochloride salt is the active ingredient and is sold under the trade name - **Benadryl**, a popular cough syrup.



Identify the true statement

A) reagent - 1 is a 3° amine

B) reagent - 3 is calcium metal accompanied with dry heat

C) reagent - 5 is NaOH and the reaction proceeds via $S_N(NGP)$ pathway

D) the active ingredient of the medicine is a molecule containing a total of 2 lone pair of electrons

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16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	745	NUMBER OF QUALIFIED RANKS	4,212

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